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Spectroscopic properties and simulation of white-light in Dy^{3+}-doped silicate glass

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ABSTRACT

Spectroscopic properties of various concentrations Dy^{3+}-doped silicate glasses were characterized by excitation and emission spectra. The optimal doping concentration of Dy^{3+} ions was found to be 3.0 wt%, and the nature of resonance energy transfer was confirmed to be electric dipole–dipole interaction according to Huang's rule. Simulation of white-light for these glasses was also performed by varying the excitation wavelength. The results show that the white-light luminescence color could be tuned to various wavelength excitations, and the present silicate glass is more suitable for generation of white-light for blue LED chips.

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1. Introduction

As a potential candidate for replacement conventional incandescent and fluorescent lamps, white-light emitting diodes (w-LEDs) has attracted intensive attention in recent years due to the advantages of long lifetime, saving energy, high efficiency, reliability, and its environmental-friendly characteristics [1–3]. At present, the common way for assemble w-LEDs is combining an ultraviolet (UV) or blue chip with down-converted phosphors [2]. The first w-LEDs was commercialized by combining a GaN-based blue chip with yellow YAG:Ce phosphors in 1996 [4]. However, the lack of red component in YAG:Ce phosphors leads to low color-redering index (CRI, Ra in the 60–75 range). As may be expected, red enhanced YAG:Ce or a small amount of red phosphors was introduced to improve the CRI to the acceptable range (Ra > 80) and increased the light conversion [5,6]. In addition to the blue LEDs plus yellow approach, three-band w-LEDs were also proposed to achieve by the combination of the blue GaN-based LED with green and red emitted phosphors or the pumping of tri-color (red, green and blue) phosphors with ultraviolet (UV) or violet LED [3]. Three-band w-LEDs maintain a very high CRI (Ra > 90) and were believed to offer the greatest potential for high efficiency solid state light [7]. For excellent CRI, both methods need efficient red phosphors that should have the excitation wavelength matching with the emission wavelength of blue LEDs (440–470 nm) or the UV/violet LEDs (350–420 nm). So far, most of these w-LEDs were limited to use as back light sources of liquid crystals displays for handy phones or digital cameras in view of low luminous flux from one w-LED [8]. Up to now, many studies have been carried out to obtain enough brightness for general lighting, in which useful method is to increase output power of LED chips. However, this also increases the chip temperature, which may cause a deterioration of the resin, which is used to fix the powder phosphors onto the LED chip, and decrease the luminous efficiency and lifetime [9].

The versatility of glasses regarding the possibility of a wide doping concentration and the narrow lines emission spectra of the lanthanide ions could be considered as a promising alternative approach since the first simulation of white-light in borate glass [10]. Compared with phosphors, glasses have more advantages such as lower production cost, simpler manufacture procedure, and free from halo effect [10–19]. On the other hand, the visible luminescence of Dy^{3+} ($4f^9$) ion mainly consists of two intense bands in the blue (470–500 nm) and yellow (570–600 nm) regions, which are associated with the $^4F_{9/2} \rightarrow {^6H_{15/2}}$ and $^4F_{9/2} \rightarrow {^6H_{15/2}}$ transitions, respectively. The latter one is a hypersensitive transition, which is strongly influenced by the environment. At a suitable yellow to blue (Y/B) intensity ratio, Dy^{3+} ions will emit white-light. Thus, luminescent materials doped/codoped with Dy^{3+} ions are usually used to the generation of white-light both in glasses [11–16] and phosphors [20–22]. These studies on glasses have emphasized on luminescent behavior and generation of white-light in Dy^{3+}-doped, Dy–Ce-, Dy–Eu-, or Dy–Tm-codoped aluminasilicate [12], borosilicate [16], phosphate glass [11] and oxyfluoride [13–15] glasses. To the best of our knowledge, little work is reported in the conventional silicate glass. In the present work, spectroscopic properties and simulation of white-light in various concentrations Dy^{3+}-doped silicate glasses were investigated in details, here silicate glass is chosen as the host glass because of its high chemical stability and low cost.

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2. Experimental

Glass samples were prepared using high-purity grade oxide or salts including SiO_2, $BaCO_3$, Al_2O_3, R_2CO_3(R = Li, Na and K) and Re_2O_3 (Re = Dy, and Y) as the starting materials. The nominal composition of all glass samples are listed in Table 1. The glass samples were all made up to contain 12.37 wt% rare-earths in total. When the concentration of Dy_2O_3 was required to less than 12.37 wt%, the remainder of the 12.37 wt% was made up by the addition of an inert oxide such as Y_2O_3. This was done in order to attempt to keep the structure of the glass matrix the same in all samples [23]. The starting materials were carefully mixed and melted at 1550 °C for 3–5 h in the normal atmosphere. After melting, the glass melts were poured into a preheated stainless steel mold for quenching and annealed at 600 °C for 3 h to release its inner stress. The glass samples with regular size of Ø20 mm × 4 mm were finally prepared after being cut and polished, and were subjected to optical measurements.

Excitation and emission spectra were recorded on a Perkin–Elmer luminescence spectrometer LS55 using a Xe lamp as an excitation source. Glass density was measured by Archimedes method using deionized water as immersion liquid. All the measures are carried out at room temperature.

3. Results and discussion

3.1. Spectroscopic properties

The excitation (monitor at 575 nm) and emission spectra (under the 349 nm excitation) of various concentrations Dy^{3+}-doped silicate glasses are shown in Figs. 1 and 2, respectively. All of the emissions in Fig. 2 are due to the 4*f*–4*f* transitions of Dy^{3+} ions. Under the excitation at 349 nm, blue-light emission peaking at 484 nm and yellow-light emission peaking at 575 nm as well as a rather weak emission peaking at 668 nm (plotted by a factor of 10 in the 640–690 nm range) were observed and can be assigned to the $^4F_{9/2} \rightarrow {}^6H_{15/2}$, $^4F_{9/2} \rightarrow {}^6H_{13/2}$ and $^4F_{9/2} \rightarrow {}^6H_{11/2}$ transitions of Dy^{3+} ions, respectively [14,23]. One can also find that the emission lines of Dy^{3+} ions are broadened somewhat because there are several Stark levels for the $^4F_{9/2}$ and ^6H_J levels. The glass sample with 3 wt% Dy^{3+} content has the strongest emission intensity. The excitation spectra of Dy^{3+} ions in Fig. 1 have the similar characteristics. Excited bands centered at 324 nm ($^6H_{15/2} \rightarrow {}^4M_{17/2}$), 349 nm ($^6H_{15/2} \rightarrow {}^4M_{15/2}$, $^6P_{7/2}$), 363 nm ($^6H_{15/2} \rightarrow {}^4I_{11/2}$), 386 nm ($^6H_{15/2} \rightarrow {}^4I_{13/2}$, $^4F_{7/2}$), 423 nm ($^6H_{15/2} \rightarrow {}^4G_{11/2}$), 451 nm ($^6H_{15/2} \rightarrow {}^4I_{15/2}$) and 472 nm ($^6H_{15/2} \rightarrow {}^4F_{9/2}$) are observed by monitoring emission at 575 nm [13]. The shape and position of these excitation peaks does not be influenced by the elevated Dy^{3+} contents, except that the 349 nm is obviously red shifted and begins to appear the Stark splitting when the concentration of Dy^{3+} ions exceeds 3.0 wt%. Furthermore, the excitation intensity ratio of 349 to 385 nm increases with the elevated Dy^{3+} concentration. It is interesting that the excitation intensity of 385 nm is higher than that of 349 nm after concentration quenching, which is contrast with the reversal results appearing in those glasses below 3.0 wt% concentrations. The abundant excitation peaks in the 350–480 nm range implies that the Dy^{3+} ions can be efficiently excited by the popular blue (440–470 nm) and the near-UV (350–420 nm) LEDs [13], it is of significance for the white-light emissions in practical application.

Table 1
Glass composition (wt%).

Glass sample	SiO_2	BaO	Al_2O_3	$Li_2O + Na_2O + K_2O$	Dy_2O_3	Y_2O_3
0.5	46.7	35.9	1.0	4.03	0.57	11.8
1.0	46.7	35.9	1.0	4.03	1.15	11.22
2.0	46.7	35.9	1.0	4.03	2.3	10.07
3.0	46.7	35.9	1.0	4.03	3.44	8.93
4.5	46.7	35.9	1.0	4.03	5.16	7.21
6.0	46.7	35.9	1.0	4.03	6.89	5.48
7.5	46.7	35.9	1.0	4.03	8.61	3.76
9.0	46.7	35.9	1.0	4.03	10.33	2.04

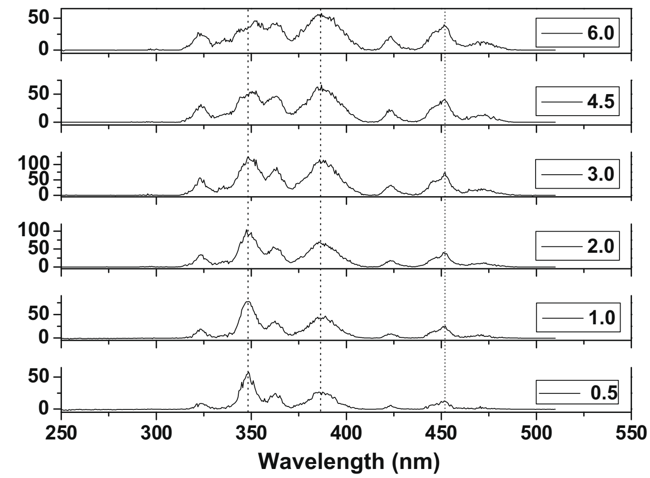

Fig. 1. Excitation spectra of Dy^{3+}-doped silicate glasses (λ_{em} = 575 nm).

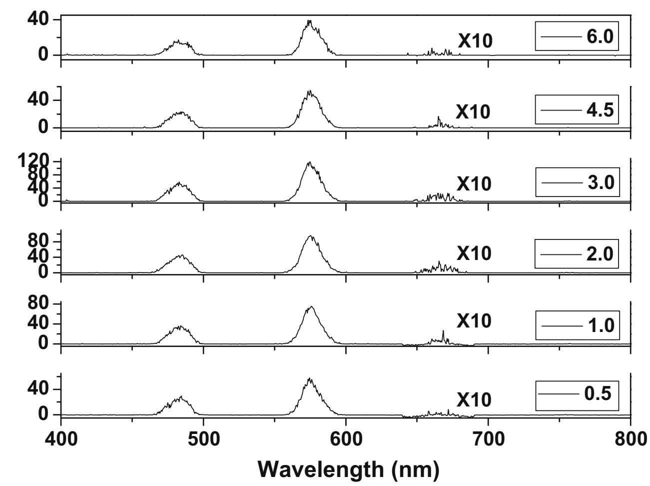

Fig. 2. Emission spectra of Dy^{3+}-doped silicate glasses (λ_{ex} = 349 nm).

The analogous emissions as Fig. 2 can also be observed by either 385 or 451 nm excitation wavelength (not shown in this work). The energy level diagram of Dy^{3+} ions and its visible transition were also illustrated in Fig. 3 based on the calculated values [24]. When the 4*f* higher energy level of Dy^{3+} ions is excited by 349 nm (or 385 and 451 nm) wavelength light, the initial population relaxes to the lower energy levels until it arrives at the $^4F_{9/2}$ level by phonon-assisted process, then it give rise to the 484, 575 and 668 nm characteristics emission. With the increasing of Dy^{3+} ions, these characteristic emissions increase firstly before quenching concentration at 3.0 wt%, and subsequently decreases dramatically exceeding this critical concentration.

3.2. Concentration quenching and resonance energy transfer

The nature of resonance energy transfer in Dy^{3+}-doped silicate and borate glasses was proved to be electric dipole–dipole interaction based on the Inokuti-Hirayama model by Nagli et al. [25] and

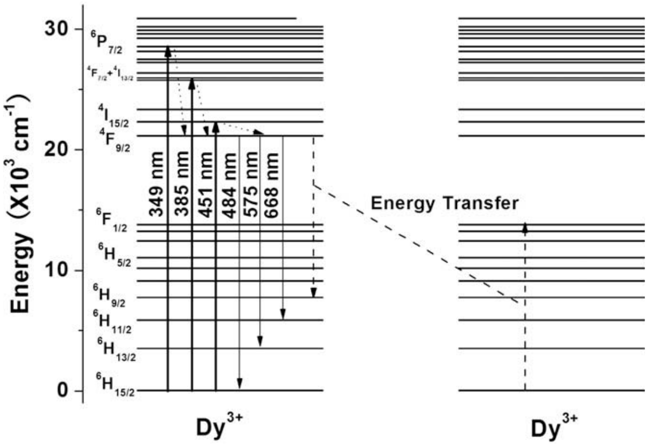

Fig. 3. Energy level diagrams, visible emission transition for Dy^{3+}, and the resonance energy transfer among Dy^{3+} ions.

Jayasankar et al. [26], respectively. In the present work, we also can determine the energy transfer type of Dy^{3+} ions in the investigated glass by Huang's rule [27]. Huang has developed a theoretical description for the relationship between the integrated intensity of luminescent ions and the corresponding molar concentration. And some recent experimental results by Meng et al. [28] and Tian et al. [29] have shown an agreement with the theoretical description. According to Huang's rule, the relationship between luminescent intensity I and doping concentration C could be expressed as

$$I \propto \alpha^{(1-\frac{s}{d})}\Gamma\left(1+\frac{s}{d}\right) \tag{1}$$

$$\alpha = C\Gamma\left(1-\frac{d}{s}\right)\left[X_0\frac{(1+A)}{\gamma}\right]^{\frac{d}{s}} \tag{2}$$

where γ is the intrinsic transition probability of sensitizer, s is index of electric multipole, which is 6, 8 and 10 for electric dipole–dipole, electric dipole–quadrupole, and electric quadrupole–quadrupole interaction, respectively. If $s = 3$, the interaction type is an exchange interaction. d is the dimension of the sample, here $d = 3$. A and X_0 are constants and $\Gamma(1 + s/d)$ is a Γ function. From Eqs. (1) and (2), it can be derived that

$$\log\left(\frac{I}{C}\right) = -\frac{s}{d}\log C + \log f \tag{3}$$

where f is independent of the doping concentration.

Fig. 4 shows the log (I/C)–log(C) plot for the $^4F_{9/2} \rightarrow {}^6H_{15/2}$ transitions of Dy^{3+} ions in the investigated silicate glass. According to Eq. (3), using linear fitting to deal with the experimental data in the region of high concentrations, the value of the slope parameter $-s/d$ was obtained to be -2.3 for the $^4F_{9/2} \rightarrow {}^6H_{15/2}$ transition. The slope parameter is approximately to -2, so the index of the electric multipole energy transfer is 6. This means that the electric dipole–dipole interaction mechanism is dominant for the energy transfer among Dy^{3+} ions in silicate glass, it is consistent with what have been reported in Refs. [25,26].

As has shown in Fig. 3, the energy gap between $^4F_{9/2}$ and $^6H_{9/2}$ levels matches well with that of $^6F_{1/2}$ and $^6H_{15/2}$ levels, it indicates that energy transfer can occur in the form of $^4F_{9/2} + {}^4F_{9/2} \rightarrow {}^6H_{9/2} + {}^6F_{1/2}$ among Dy^{3+} ions by electric dipole–dipole interaction. The probability of this energy transfer gets higher with the elevated Dy^{3+} concentration, particularly in high concentrated Dy^{3+}-doped glass, which leads to the depopulation of the $^4F_{9/2}$ level and results in the decrease of the 484 and 575 nm emissions.

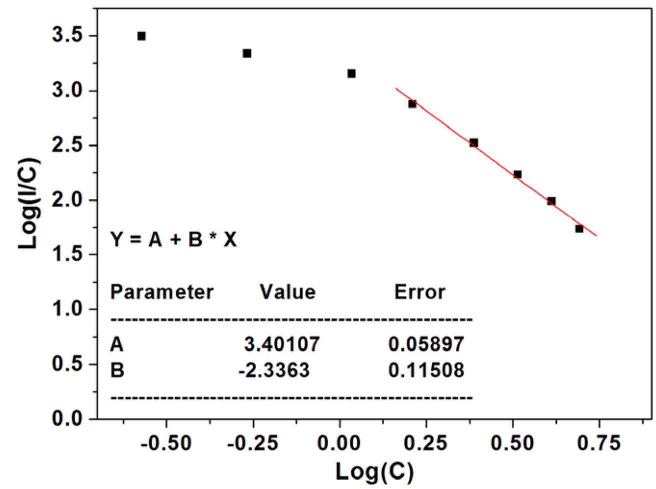

Fig. 4. The relation of the concentration of Eu^{3+} ions log(C) and the log(I/C) for the $^4F_{9/2} \rightarrow {}^6H_{15/2}$ transition by 349 nm excitation.

3.3. Simulation of white-light

From the emission spectra shown in Fig. 2, the integrated intensity ratios of yellow to blue (Y/B) are calculated and presented in Table 2. As can be seen, the ratios vary from 1.959 to 2.816 with the elevated Dy^{3+} concentration except the 2.0 wt% one. The change in the Y/B ratio is attributed to the change in the environment of Dy^{3+} ions in materials as it involves a hypersensitive transition $^4F_{9/2} \rightarrow {}^6H_{13/2}$ with $\Delta J = 2$. It is worth to note that the Y/B ratios changes a little before concentration quenching, but it increases noticeably when the Dy^{3+} contents exceeds the critical 3.0 wt%. So it was concluded that the environment of Dy^{3+} ions may be much changed due to the clustering of Dy^{3+} ions in glass after concentration quenching.

The Y/B intensity ratios of visible emissions increases over a wide range indicating the feasibility of generation of with-light in silicate glass. The chromaticity color coordinates of all the investigated glasses were calculated and shown in Table 2, and the corresponding Commission International de I'Eclairage (CIE) 1931 $x - y$ chromaticity diagram are also presented in Fig. 5. The CIE coordinates of all investigated glasses lies within the white region, though they are far away from the ideal equal energy white-light illumination (0.333, 0.333) and there exists a tendency to be away from it with the elevated concentrations, as labeled along the arrow direction. Further studies on simulation of white-light for these Dy^{3+}-doped silicate glasses are necessary by both adjusting the glass compositions and enhancing the red emission part.

As mentioned above, the abundant excitation peaks of Dy^{3+} ions in the range of 350–470 nm match well with the commercial blue

Table 2
Yellow to blue integrated intensity ratios, chromaticity color coordinations of varied concentration Dy^3-doped silicate glasses by 349 nm excitation.

Glass sample	Y/B ratio	Color coordination	
		x	y
0.5	1.959	0.381	0.434
1.0	2.027	0.385	0.437
2.0	1.954	0.381	0.431
3.0	2.007	0.383	0.434
4.5	2.139	0.387	0.441
6.0	2.148	0.387	0.436
7.5	2.419	0.395	0.456
9.0	2.816	0.400	0.471

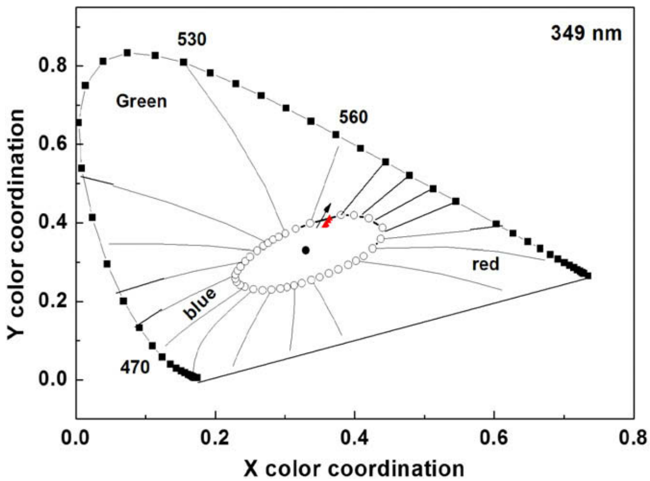

Fig. 5. CIE coordinate diagram of various Dy^{3+}-doped silicate gasses by 349 nm excitation.

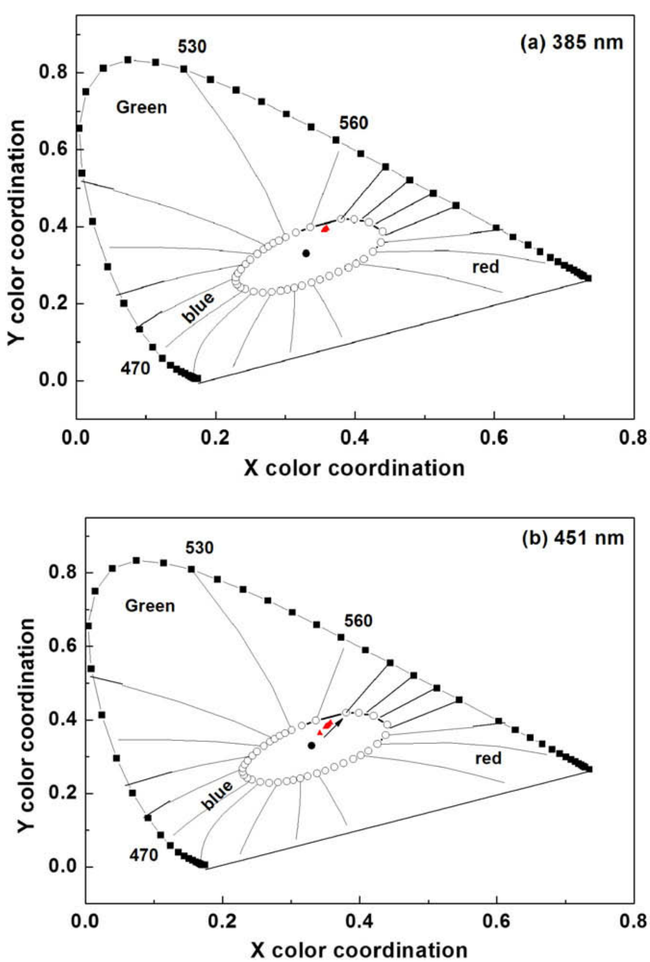

Fig. 6. CIE coordinate diagrams of various Dy^{3+}-doped silicate gasses by 358 nm (a) and 451 nm (b) excitation.

LEDs (440–470 nm) or the UV/violet (350–420 nm), and it is essential to understand the dependence of various excitation wavelengths on simulation of white-light. The representative CIE 1931 chromaticity diagrams were shown in Fig. 6 by 385 and 451 nm excitation, respectively. As shown in Fig 6(b), the observed results by 451 nm excitation was similar to that by 349 nm excitation, but the chromaticity color coordinates under 451 nm excitation is closer to the equal energy point (0.333, 0.333). The results indicate that the present glass is more suitable for the generation of white-light for the blue LED chip. However, the chromaticity color coordinates by 385 nm excitation tends to be one point far from the ideal equal energy point, as shown in Fig. 6(a).

4. Conclusions

Various concentrations Dy^{3+}-doped silicate glasses were prepared and characterized. The glass samples emit simultaneously visible blue and yellow lights as well as a weaker red emission by 349, 385 and 451 nm excitation, respectively. The quenching concentration was found to be 3.0 wt% and the electric dipole–dipole interaction mechanism was confirmed by Huang's rule. With the elevated Dy^{3+} ions concentration, the Y/B intensity ratios of visible emissions increases over a wide range indicate the feasibility of simulation of with-light in silicate glass. The simulation results demonstrate the present silicate glass is more suitable for generation of white-light for the blue LED chips.

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